

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int a[100], n, i;
6
7      printf("Enter number of elements: ");
8      scanf("%d", &n);
9
10     printf("Enter array elements:\n");
11     for(i = 0; i < n; i++)
12         scanf("%d", &a[i]);
13
14     printf("Array elements are:\n");
15     for(i = 0; i < n; i++)
16         printf("%d ", a[i]);
17
18     return 0;
19 }
```

Enter number of elements: 5  
Enter array elements:  
4 5 6 7 3  
Array elements are:  
4 5 6 7 3  
...Program finished with exit code 0

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int a[100], n, i, sum = 0;
6      float average;
7
8      printf("Enter number of elements: ");
9      scanf("%d", &n);
10
11     printf("Enter array elements:\n");
12     for(i = 0; i < n; i++)
13     {
14         scanf("%d", &a[i]);
15         sum = sum + a[i];
16     }
17
18     average = (float)sum / n;
19
20     printf("Sum = %d\n", sum);
21     printf("Average = %.2f", average);
22
```



```
Enter number of elements: 6
Enter array elements:
1 2 3 4 5 6
Sum = 21
Average = 3.50
```

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int a[100], n, i, largest, smallest;
6
7      printf("Enter number of elements: ");
8      scanf("%d", &n);
9
10     printf("Enter array elements:\n");
11     for(i = 0; i < n; i++)
12         scanf("%d", &a[i]);
13
14     largest = smallest = a[0];
15
16     for(i = 1; i < n; i++)
17     {
18         if(a[i] > largest)
19             largest = a[i];
20
21         if(a[i] < smallest)
22             smallest = a[i];

```



input

```

Enter number of elements: 6
Enter array elements:
2 3 4 7 4 2
Largest = 7
Smallest = 2

```

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int a[100], n, i;
6      int even = 0, odd = 0;
7
8      printf("Enter number of elements: ");
9      scanf("%d", &n);
10
11     printf("Enter array elements:\n");
12     for(i = 0; i < n; i++)
13     {
14         scanf("%d", &a[i]);
15
16         if(a[i] % 2 == 0)
17             even++;
18         else
19             odd++;
20     }
21
22     printf("Even elements = %d\n", even);

```



```

Enter array elements:
2 3 4 6 7
Even elements = 3
Odd elements = 2

```

input

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int a[100], n, i;
6
7      printf("Enter number of elements: ");
8      scanf("%d", &n);
9
10     printf("Enter array elements:\n");
11     for(i = 0; i < n; i++)
12         scanf("%d", &a[i]);
13
14     printf("Reversed array:\n");
15     for(i = n - 1; i >= 0; i--)
16         printf("%d ", a[i]);
17
18     return 0;
19 }

```

input

```

Enter number of elements: 6
Enter array elements:
3 4 5 6 7 8
Reversed array:
8 7 6 5 4 3

```

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int a[100], n, i, pos, value;
6
7      printf("Enter number of elements: ");
8      scanf("%d", &n);
9
10     printf("Enter array elements:\n");
11     for(i = 0; i < n; i++)
12         scanf("%d", &a[i]);
13
14     printf("Enter position to insert: ");
15     scanf("%d", &pos);
16
17     printf("Enter element: ");
18     scanf("%d", &value);
19
20     for(i = n; i >= pos; i--)
21         a[i] = a[i - 1];
22

```

input

```

Enter number of elements: 4
Enter array elements:
2 44 55 22
Enter position to insert: 3
Enter element: 1
Array after insertion:
2 44 1 55 22

```

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int a[100], n, i, pos;
6
7      printf("Enter number of elements: ");
8      scanf("%d", &n);
9
10     printf("Enter array elements:\n");
11     for(i = 0; i < n; i++)
12         scanf("%d", &a[i]);
13
14     printf("Enter position to delete: ");
15     scanf("%d", &pos);
16
17     for(i = pos - 1; i < n - 1; i++)
18         a[i] = a[i + 1];
19
20     n--;
21
22     printf("Array after deletion:\n");

```

input

```

Enter number of elements: 5
Enter array elements:
3 4 5 2 3
Enter position to delete: 2
Array after deletion:
3 5 2 3

```

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int a[50], b[50], c[100];
6      int n1, n2, i;
7
8      printf("Enter number of elements in first array: ");
9      scanf("%d", &n1);
10
11     printf("Enter first array elements:\n");
12     for(i = 0; i < n1; i++)
13         scanf("%d", &a[i]);
14
15     printf("Enter number of elements in second array: ");
16     scanf("%d", &n2);
17
18     printf("Enter second array elements:\n");
19     for(i = 0; i < n2; i++)
20         scanf("%d", &b[i]);
21
22     for(i = 0; i < n1; i++)

```

input

```

Enter first array elements:
4 5 6 7 8
Enter number of elements in second array: 4
Enter second array elements:
1 2 3 4
Merged array:
4 5 6 7 8 1 2 3 4

```

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[100], b[100], n, i;
6
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    printf("Enter array elements:\n");
11    for(i = 0; i < n; i++)
12        scanf("%d", &a[i]);
13
14    for(i = 0; i < n; i++)
15        b[i] = a[i];
16
17    printf("Copied array:\n");
18    for(i = 0; i < n; i++)
19        printf("%d ", b[i]);
20
21    return 0;
22 }
```

input

```
Enter number of elements: 5
Enter array elements:
2 3 4 5 6
Copied array:
2 3 4 5 6
```

```

1 #include <stdio.h>
2
3 int main()
4 {
5     int a[100], n, i, j;
6
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    printf("Enter array elements:\n");
11    for(i = 0; i < n; i++)
12        scanf("%d", &a[i]);
13
14    printf("Duplicate elements:\n");
15
16    for(i = 0; i < n; i++)
17    {
18        for(j = i + 1; j < n; j++)
19        {
20            if(a[i] == a[j])
21            {
22                printf("%d ", a[i]);

```

input

```

Enter number of elements: 7
Enter array elements:
87 5 5 7 8
7 7
Duplicate elements:
5 7 7

```